

Problem-Solution Fit

Date	03 July 2026
Team ID	N/A
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

The canvas below maps how OptiCrop's solution addresses the root causes, behaviors, and limitations of its target users (farmers, researchers, and policymakers).

1. CUSTOMER SEGMENT(S) [CS] Smallholder and marginal farmers Agricultural researchers and policymakers Agribusinesses and agri-extension organizations	6. CUSTOMER LIMITATIONS [CL] Limited internet access in rural areas Low-cost shared smartphones/PCs Limited digital literacy Limited budget for soil testing	5. AVAILABLE SOLUTIONS [AS] Manual agronomist consultation — accurate but slow, costly, doesn't scale Tradition/experience-based crop choice — fast but not adaptive Generic government advisories — broad, not personalized
2. PROBLEMS / PAINS [PR] Choosing an unsuitable crop for current soil/climate — recurring every planting season Inefficient fertilizer/water use from guesswork Slow manual analysis of soil reports across regions	9. PROBLEM ROOT / CAUSE [RC] Crop suitability depends on many interacting variables (N, P, K, temperature, humidity, pH, rainfall) that are hard to assess manually and consistently at scale	7. BEHAVIOR [BE] Farmers rely on local agronomists or neighbors' choices (moderate-high reliance) Researchers manually cross-reference spreadsheets/reports (high effort, high stakes)
3. TRIGGERS TO ACT [TR] Start of a new planting/sowing season Declining yields New soil test results Policy planning cycles	10. YOUR SOLUTION [SL] OptiCrop: an ML-based web app that takes soil nutrient and climate inputs and instantly recommends the most suitable crop, backed by a Random Forest model chosen after comparing multiple algorithms.	8. CHANNELS OF BEHAVIOR [CH] ONLINE: web browser (smartphone/PC), WhatsApp/SMS groups OFFLINE: agri-extension office visits, cooperative meetings, local seed shops
4. EMOTIONS [EM] Before: anxious, uncertain, financially at risk After: confident, reassured, more in control of the decision		